DOUBLE JEOPARDY: Declarations

DOUBLE JEOPARDY: The Vulnerability Spiral of Compound Climate Risk Across 5 Islands

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Preprint of the first version: EarthArXiv, DOI 10.31223/X5M50Q · Code and data: github.com/sakshimaske303-commits/DOUBLE-JEOPARDY

Highlights

- On a bare-earth elevation model, 11.8% of people in the Maldives and 6.4% in Lakshadweep live at or below 1 m, against 8.9% and 0.8% on a surface model.
- Reef heat stress was higher in 2016–2020 than in 1996–2000 on all four reef islands, but no trend is significant once serial correlation is accounted for; mangrove extent shows no decline.
- Seychelles and the Maldives are in a near tie on the compound score, and the top island changes at an exposure weight of 52%.
- Protection of the 10 km coastal zone, from 1.3% to 71.4%, does not follow the compound score (descriptive, n = 4).
- National coastal setbacks reach fewer than one in ten low-lying buildings on the four tropical island groups, against 85% under the Canary Islands' 100 m line.
- Coding the five governments' coastal rules shows that more instruments did not mean more reach: Lakshadweep has five of six instrument types and its setback reaches 7% of its low buildings; of 23 rule changes since 1988, three relaxed protection, in Spain (2013) and the Maldives (2024–2025).

Contributions, funding and interests

I am the sole author. I conceived the study, designed the methods, wrote the code, ran and checked the analysis, curated the data, wrote the paper and made the figures and maps (CRediT: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization). The work was not funded by anyone; I paid its costs myself. I have no competing interests to declare. No human participants, animals or personal data were involved, so ethical approval was not needed.

Data and their licences

The study uses public data only: DeltaDTM v1.1 and the Copernicus DEM GLO-30 for elevation, WorldPop 2020 for population, OpenStreetMap for named places and coastlines, Overture Maps building footprints, NOAA Coral Reef Watch Degree Heating Weeks, Global Mangrove Watch v3, UNEP-WCMC reef and mangrove layers, the World Database on Protected Areas (June 2026 release), Sentinel-2, the V-Dem electoral democracy index (v16, via Our World in Data) and the published texts of national coastal and environmental laws. My own code and text are released under CC BY 4.0, but third-party data keep their original licences. WDPA extracts are not included because their terms do not allow public redistribution, and the UNEP-WCMC layers are for non-commercial research use. Full citations, licences and attribution texts are in `DATA_SOURCES_AND_LICENSES.md`.

Reproducing the results

Every reported number is written to a CSV in `data/`, and the figures and dashboard read those files. `scripts/run_all.py` re-runs the analysis, figures, maps and dashboard data and ends by reporting a complete or partial run. Large rasters and the building footprints are not committed; they are rebuilt with the included download scripts. The DeltaDTM tiles are exported with the Earth Engine Python API, which needs an Earth Engine project; the rest runs without it. Two scripts recompute key results independently: the coral trend p-values without the pyMannKendall library, and the setback distances on the WGS84 ellipsoid.

Limits of the design

These are elevation thresholds, not flood models, and DeltaDTM's vertical error of about 0.4–0.5 m is large next to a 1 m threshold, so the comparison between models and islands matters more than any single percentage. Five island groups, and four for governance, describe this sample; they cannot establish general relationships. The setback analysis compares buildings with the setback written in national law, measured to a mapped shoreline rather than the legal reference line, and says nothing about permits, local plans or compliance.

What changed from the first version

Six results of the first version were withdrawn on review. The 78.3% of Seychelles settlements "at risk" came from counting the elevation raster's NoData value as land at or below 1 m. The significant coral trends came from a single 5 km cell and a test that treated months as independent. Seychelles ranked first at every weighting only because both inputs happened to be highest there. The governance correlation of r = 0.965 came from counting overlapping protected areas twice and dividing by land area; on the coastal zone it is r = −0.10. The built-up rise in Seychelles was measured over boxes that were mostly ocean. And the first version used a surface elevation model only. Each correction is recorded in `paper/DJ_Development_Log.md`.

AI assistance

The questions, the choice of islands and data, the methods, the decision to withdraw the six first-version results and the interpretation are mine. Within that work I used Claude (Anthropic), an AI assistant, in a supporting role. Its main job was cross-checking: comparing the exposure shares, reef heat-stress figures, compound scores and setback coverage quoted in the paper, executive summary, README and dashboard against the CSV files my scripts produce. I also asked it for help when one of my Python scripts broke or gave output I could not account for, and for the layout of the project web pages. The paper and other documents are my own writing; I used the assistant to flag wording that was unclear or overly technical, gaps where something seemed to be missing, and punctuation, and I decided which of those points to act on. It did not acquire or process the elevation, population, building, reef or mangrove data, and it did not run the analysis. I reviewed everything it suggested before using it, and the responsibility for the content is mine.

Licence and citation

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